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HV Electrical Systems for Vehicles

Thomas Bradley
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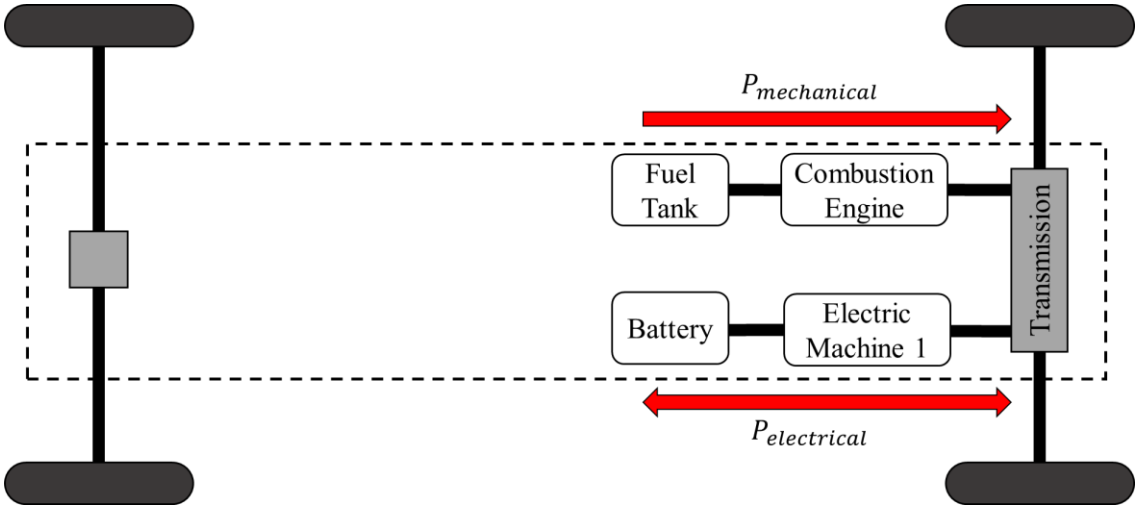
Spring, 2024

- Recall the basics of parallel HEV operation…



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- Recall the basics of parallel HEV operation…

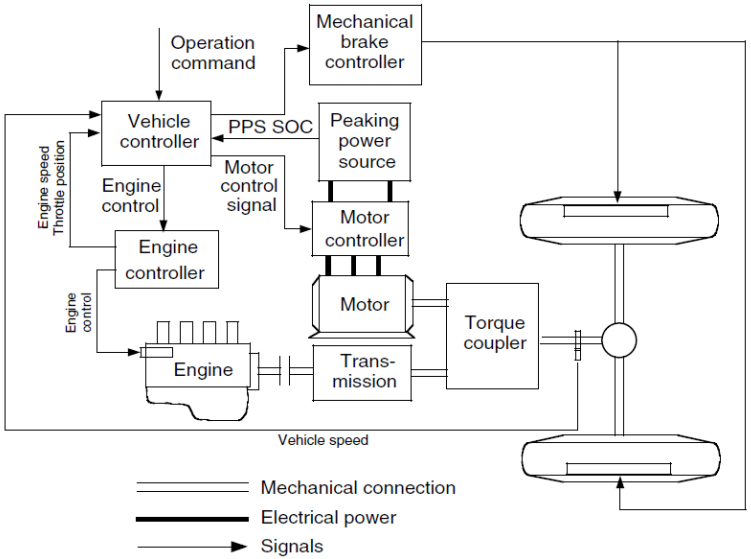
Parallel Hybrid Advantages

- Electric motor can double as a generator
  - No generator required
- Smaller components
- Fewer energy conversion efficiency penalties
- More energy captured during regenerative braking

Parallel Hybrid Disadvantages

- Control strategies governing two power sources are more complex
- Drivetrain components are more complex
- Larger engine required
- Engine cannot always operate at peak efficiency
  - Engine speed coupled to wheel speed

- Lets think about modeling this architecture…



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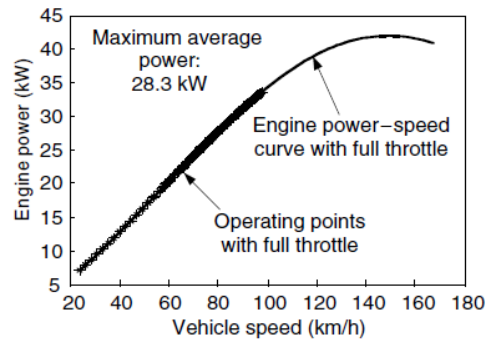
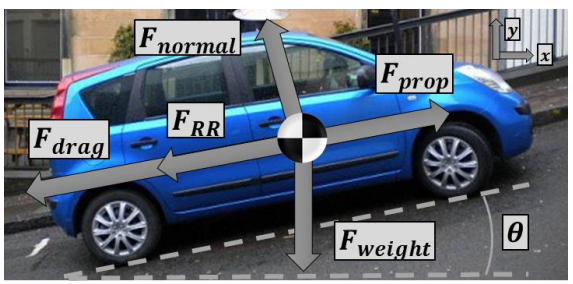
- Example: Parallelizing a conventional vehicle
  - Honda does this: 2015 Honda Civic
    - 1.5L V-TEC SOHC 4 Stroke Engine
    - Integrated Motor Assist
    - 1280 kg Curb Weight
  - Similar to the conventional version

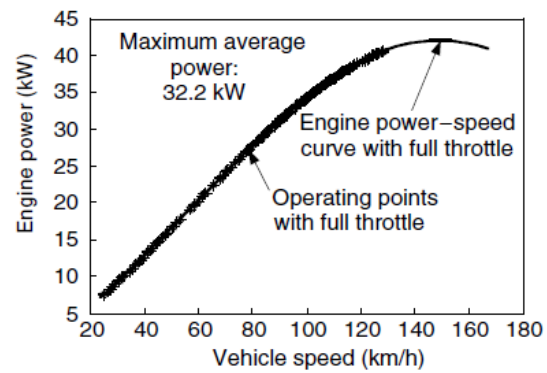
- Internal combustion engine
  - Larger than for series architecture
  - Support vehicle at normal constant speeds
  - But, engine speed = *f* (vehicle speed)
    - Not as easy to ensure proper engine operation
  - Choose initial engine size, evaluate performance



(b) FTP 75 highway driving cycle

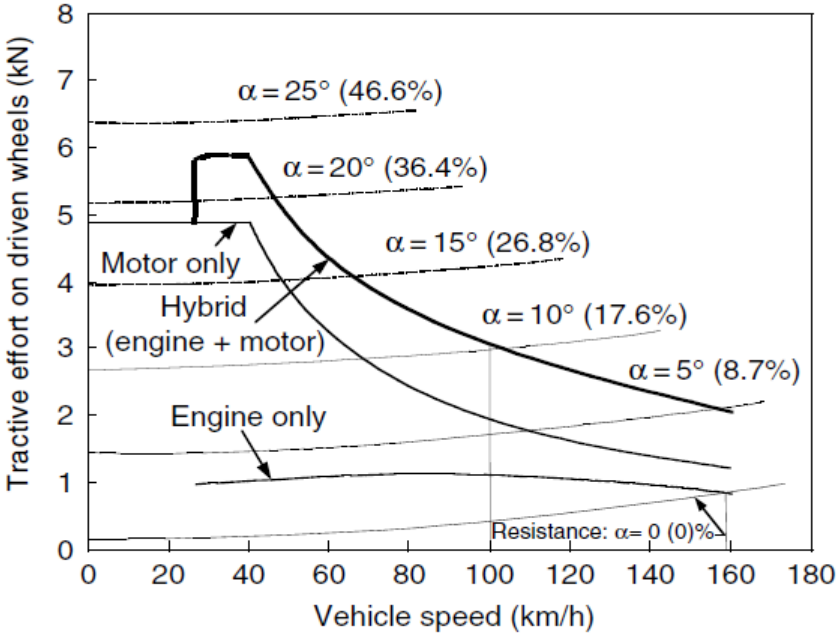


$$P_e = \frac{V}{1000\eta_{t,e}}\left(M_v g f_r + \frac{1}{2}\rho_a C_D A_f V^2 + M_v g\, i\right)(\text{kW})$$



(c) US06 driving cycle

- Electric motor
  - Can assist in during high power demand
  - Provide power during accelerations
    - Engine power for drag and rolling resistance
    - EM power during accelerations
    - Run simulations



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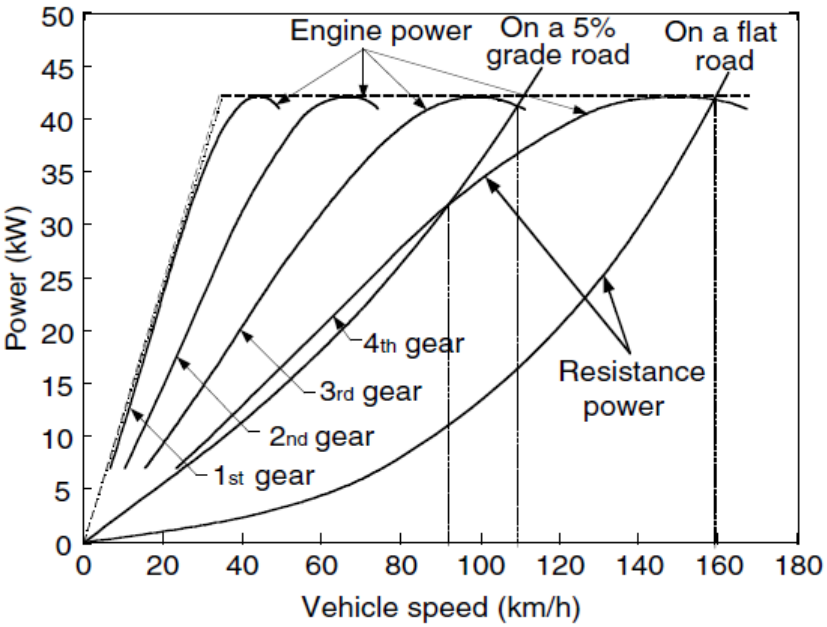
- Transmission
  - 1st gear
    - Ensure hill climbing is possible
    - Ensure acceleration power demands are possible
  - Other gears
    - Higher overall engine efficiency
    - Improved vehicle performance
    - Improved battery charging
    - Improved fuel economy
    - But: Heavier and more difficult to control



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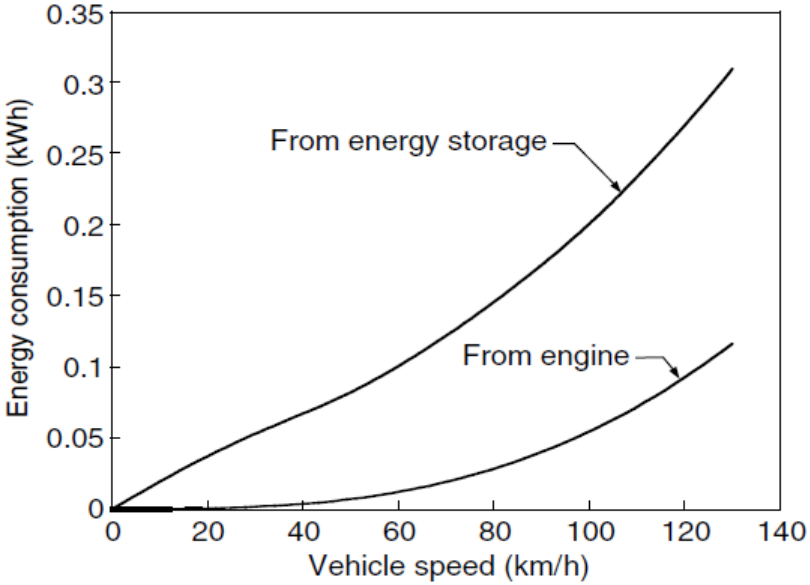
- Battery size
  - Meet high power demands
  - Assist with stop and go driving
  - Typically want to keep the state of charge high



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- Parallel HEVs
  - Can significantly improve a conventional vehicle with a few modifications
  - The control strategies can be difficult
  - Engine operation is not as efficient
  - Balance engine, EM, battery, and transmission
    - Driving power demands are met
    - Engine is sized down from a conventional vehicle
    - EM assists with high power demand
    - EM provides power during stop and go
      - Fuel consumption reduction

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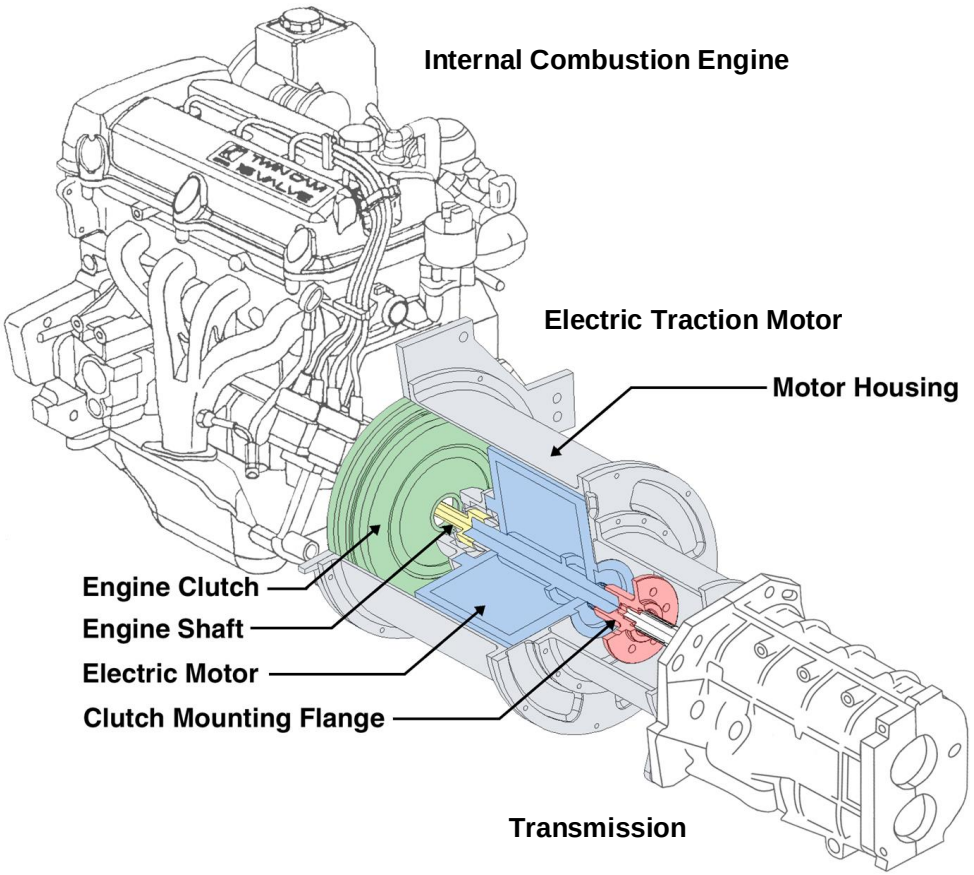
Hybrid Powertrain Architecture Parallel Configuration, Pre-transmission E-motor (P2)

Powertrain Attributes

- Capable of launching vehicle and driving in a pure-electric mode
- Engine can de-clutch/disengage from powertrain under most conditions
- E-motor is always connected, delivers power to transmission input shaft
- Transmission can be any type
  - Manual gearbox
  - Automatic
  - CVT



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- Once you have an initial specification of the vehicle parameters from your 1st order calculation, you can begin to do more detailed simulations
- These simulation show dynamics that are not in our 1st order calc including
  - Battery limitations, shift shocks, the effect of different modes, engine start, wheel slip, etc.

Parallel Powertrain Architecture

Combustion Engine

SI: M111 2.3L – 105 kW

CI: OM611 2.2L – 95 kW

Automated Clutch

E-motor

Sachs ZF DC PM

60-80 kW/240-290 Nm

Auto Trans (5-spd)

Battery

VARTA NiMH – 360V/40Ah

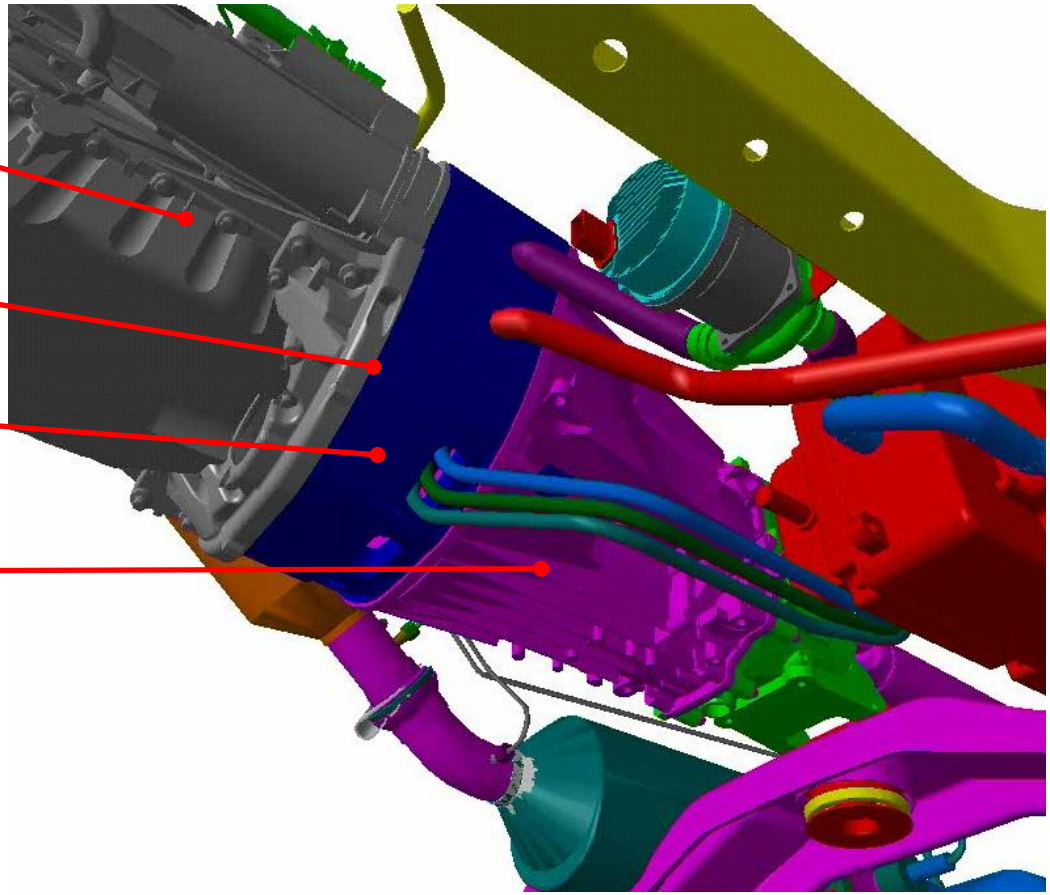
14 kWh, 85 kW

Lithium Ion – 380V/39 Ah

15 kWh, 100 kW

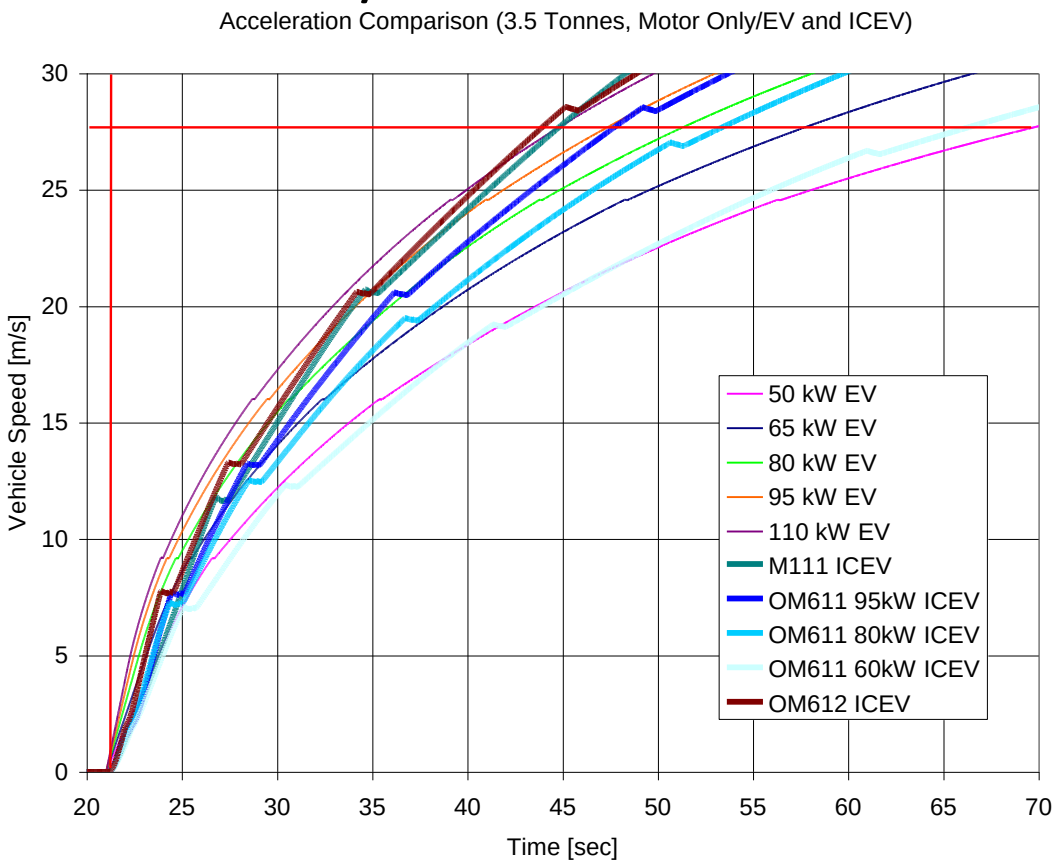
Charger

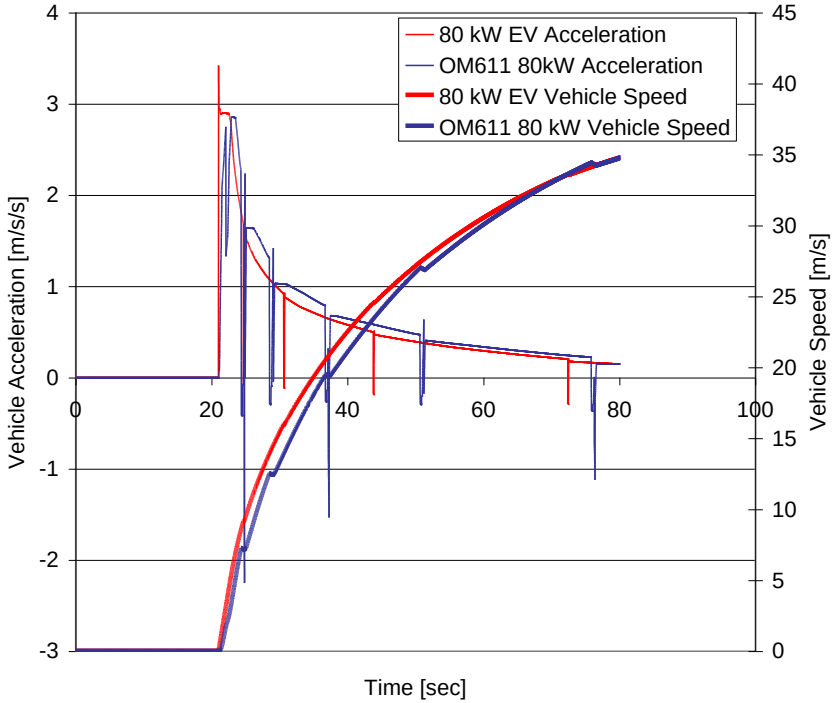
3.3 kW Conductive



Evaluation of Proposed EV/HEV Concepts

- Acceleration Performance Simulation
  - EV / ICEV Acceleration Comparisons





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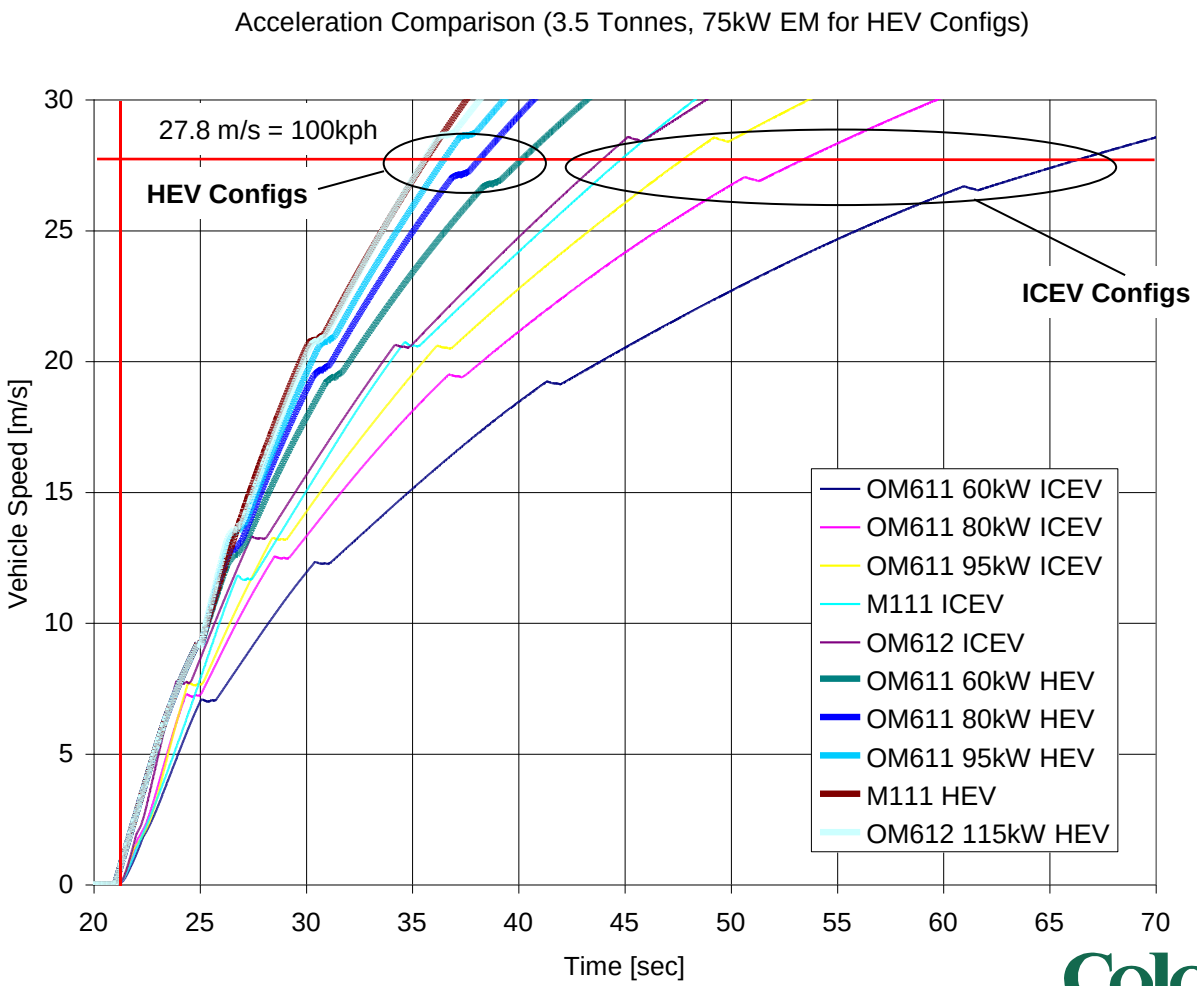
Evaluation of Proposed EV/HEV Concepts

- Acceleration Performance Simulation



- EV / ICEV Acceleration Parity is possible
- <23 sec 0-100kph (0-27.8 m/s) is preferable and possible using HEV accelerations
- EPRI Vehicle (-) ; Green Zone Vehicle (-) ; Charge Depletion Vehicle (+)

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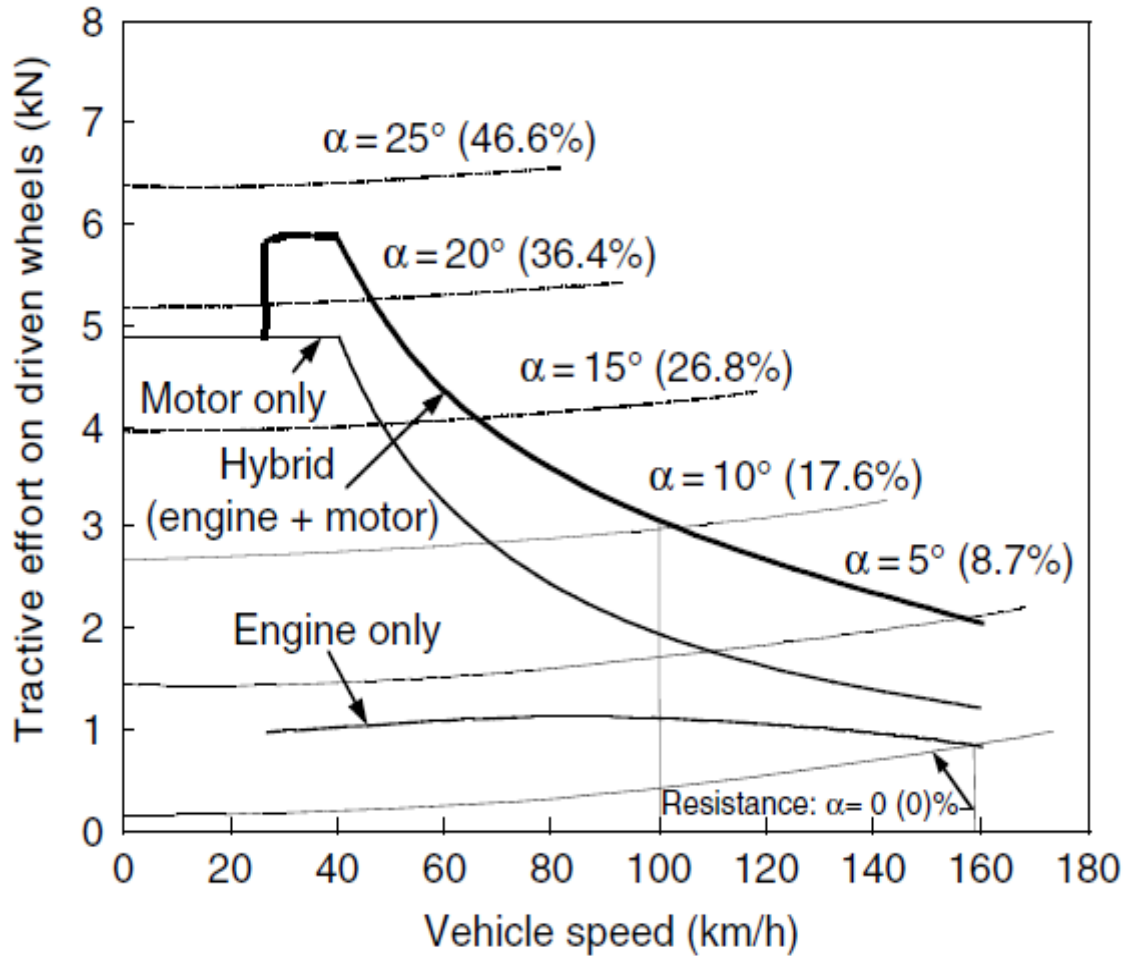
- This diagram from the book shows that in parallel vehicles, the torque available at the wheels is the sum of engine and motor torque.



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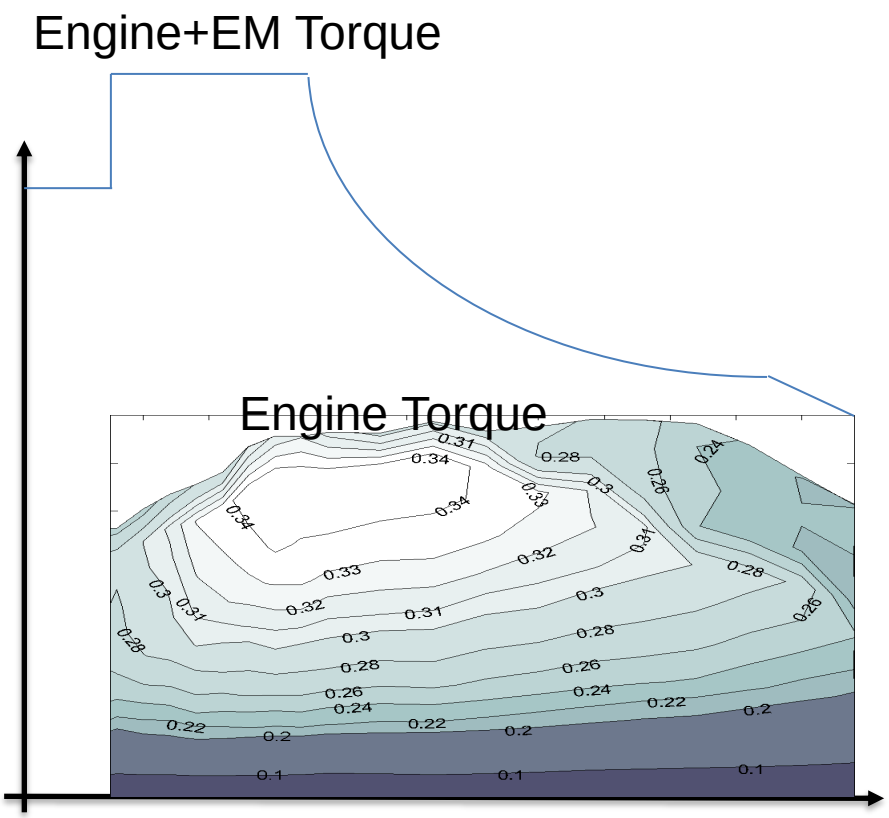

- For the more-electric HEV that has been the subject of our vehicle simulations to date, the EM and engine are more equivalently sized
- Engine Torque + EM torque yields the total powertrain torque

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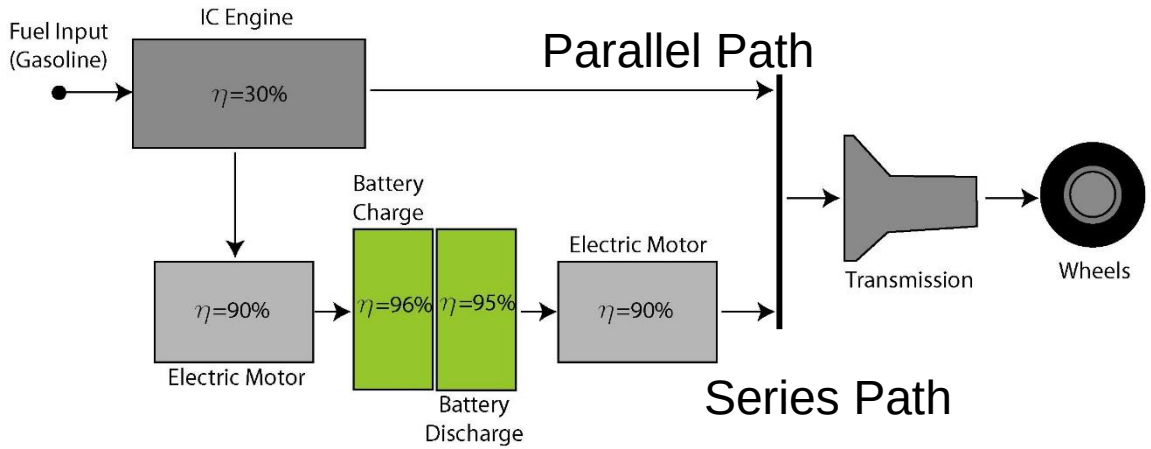
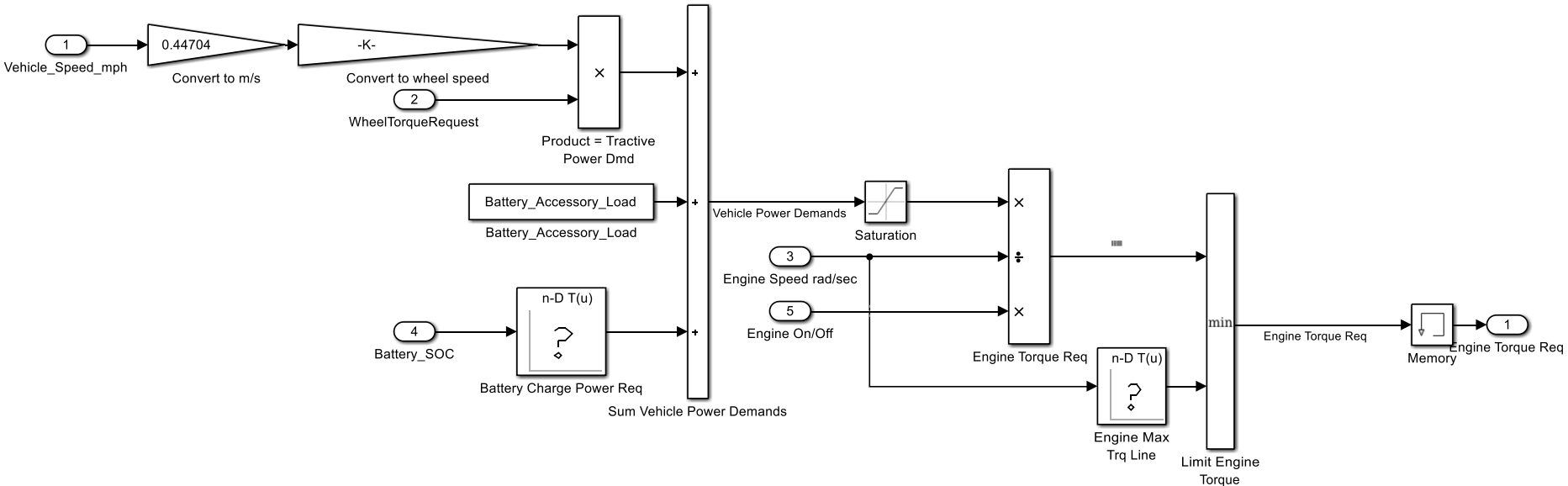
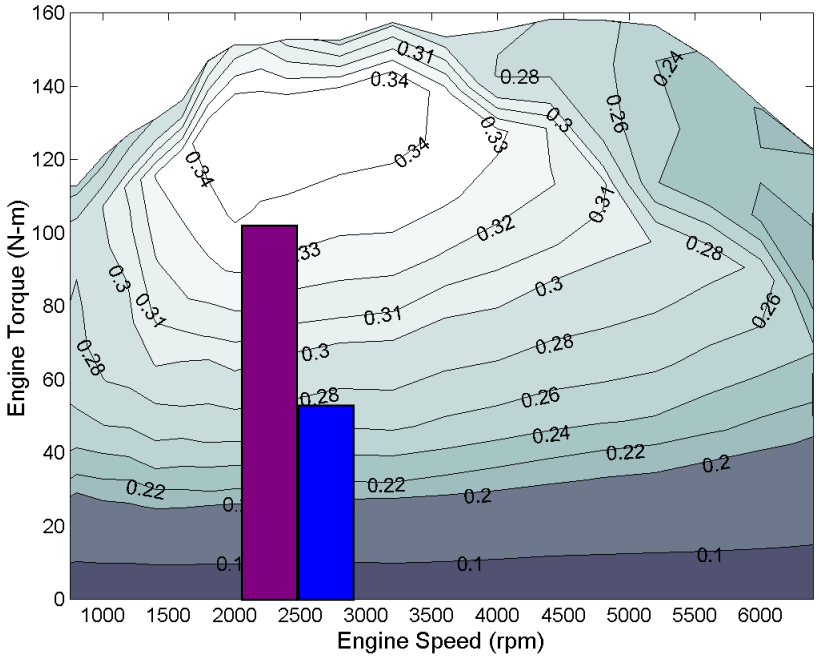
Parallel Path Control

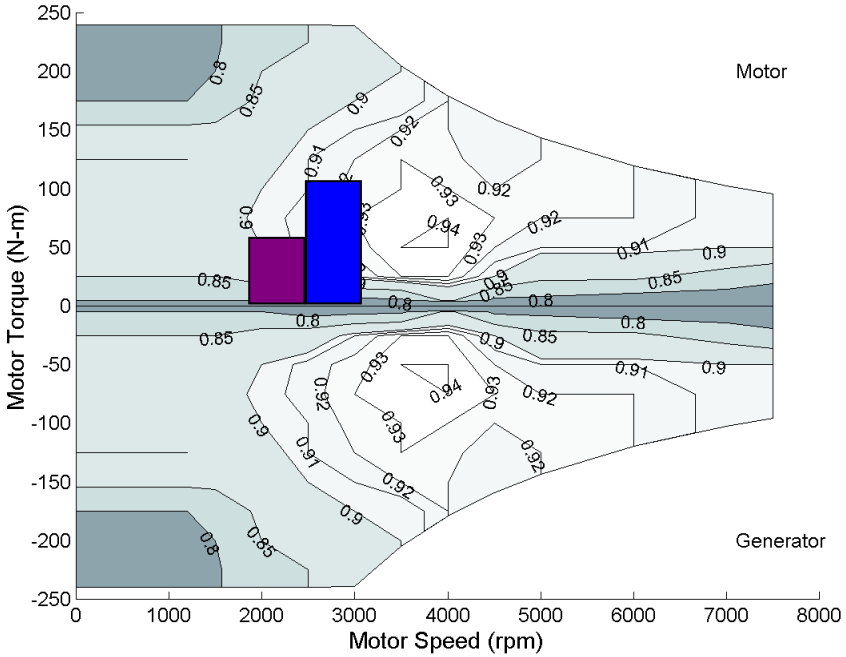




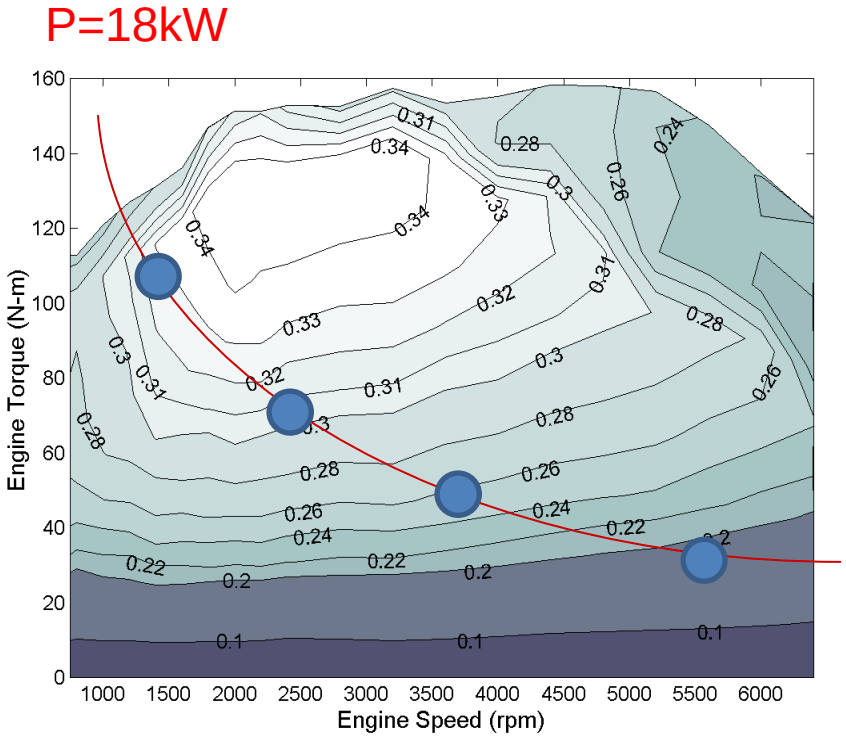
Vehicle Fuel Economy Analyses

- Torque Control
  - Charge Depleting / Charge Sustaining
    - Engine + Motor = Powertrain
    - 100 Nm + 50 Nm = 150 Nm
    - 50 Nm + 100 Nm = 150 Nm
  - Component Efficiencies





Optimal Engine Operation



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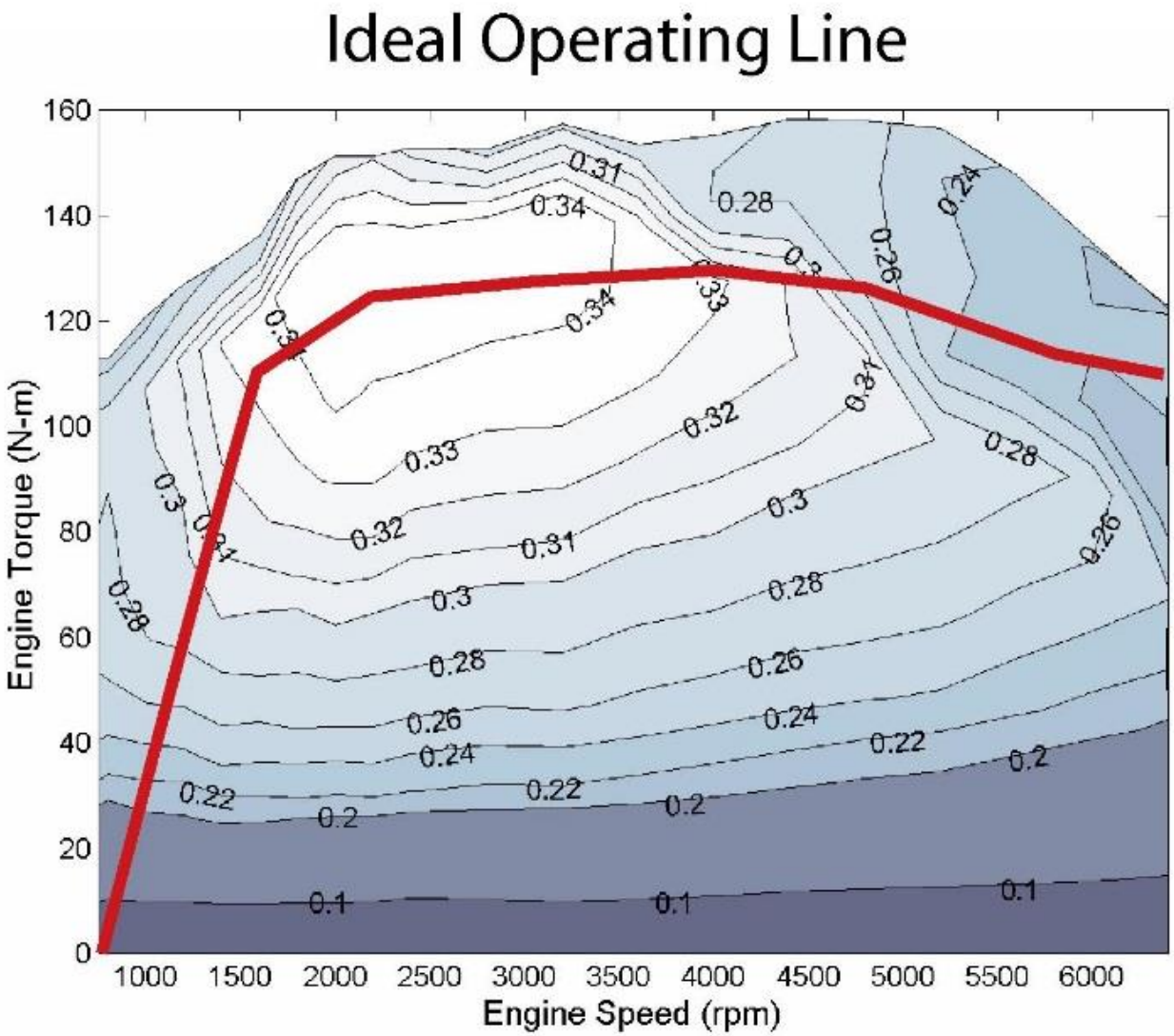
With a CVT, you can operate at the IOL



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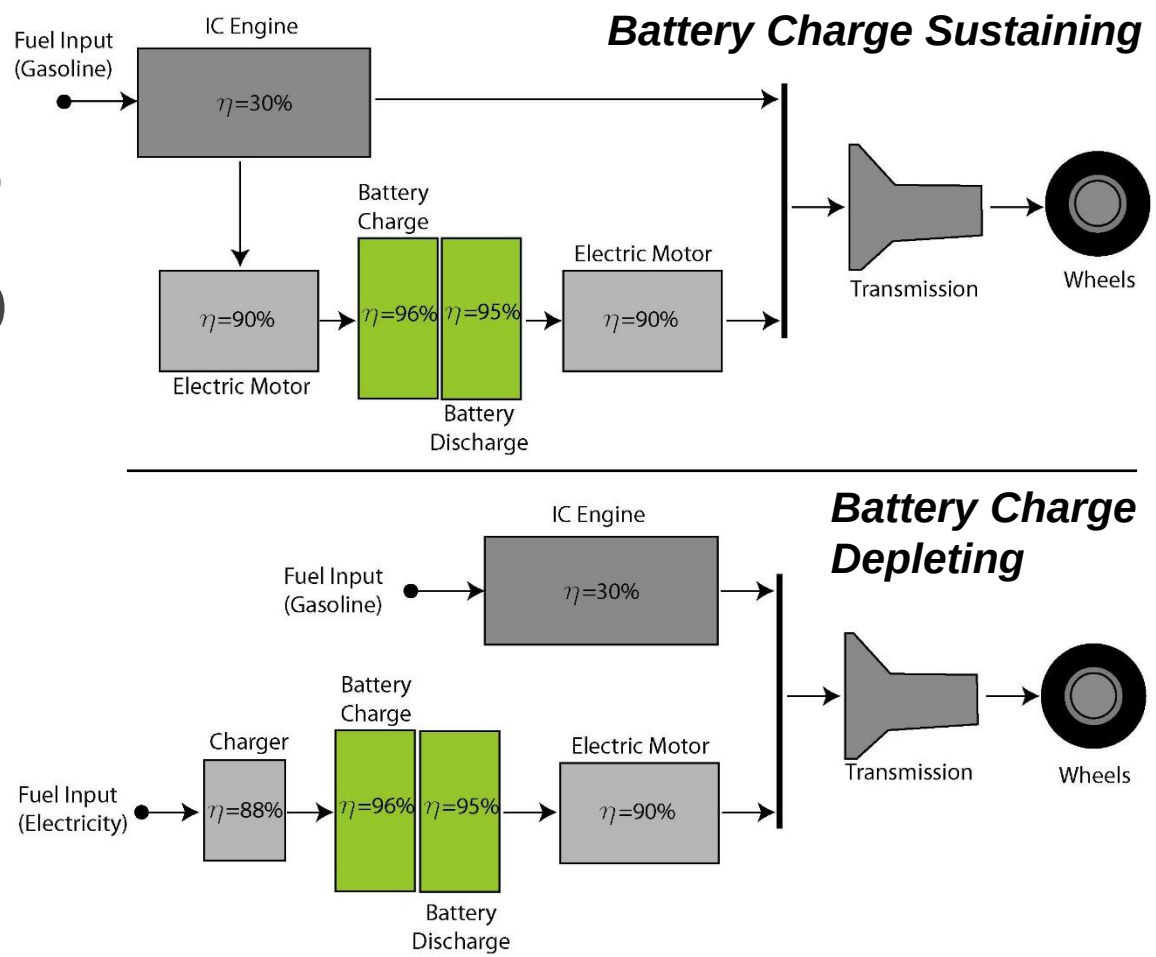
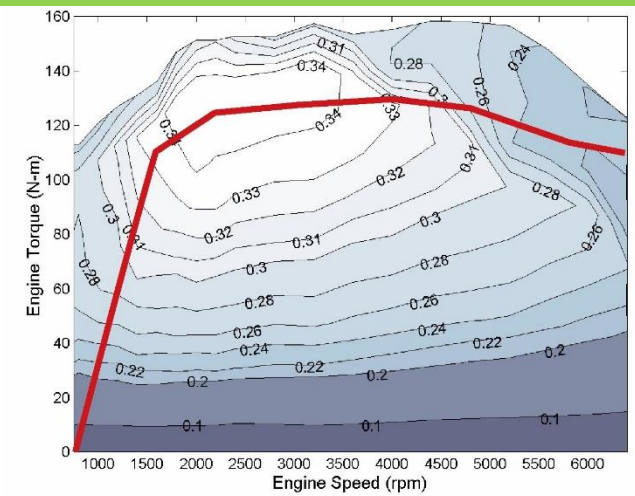
Vehicle Fuel Economy – Torque Control

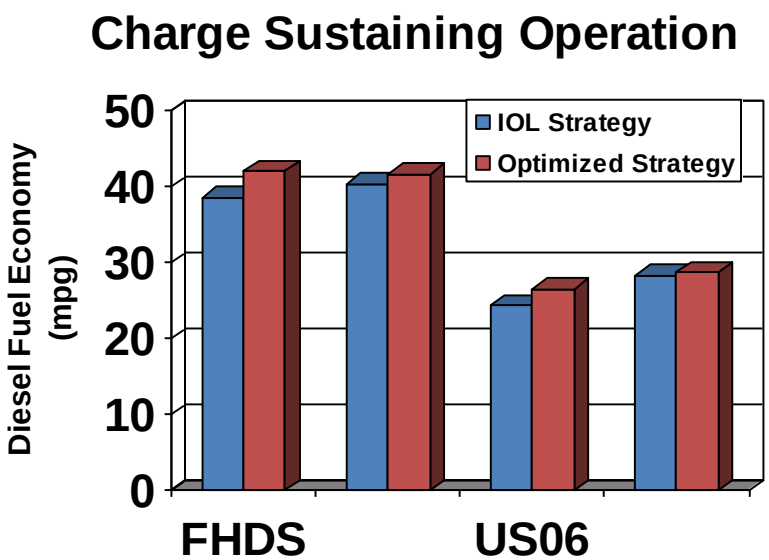
- Component efficiencies from modeling (FUDS)
- Fuel inputs need Gasoline to Electricity Equivalence:
  - Fuel Costs ($/gal, $/kWh)
  - Well to Tank Efficiency
  - CO_2 Emissions (g/kWh, g/gal)

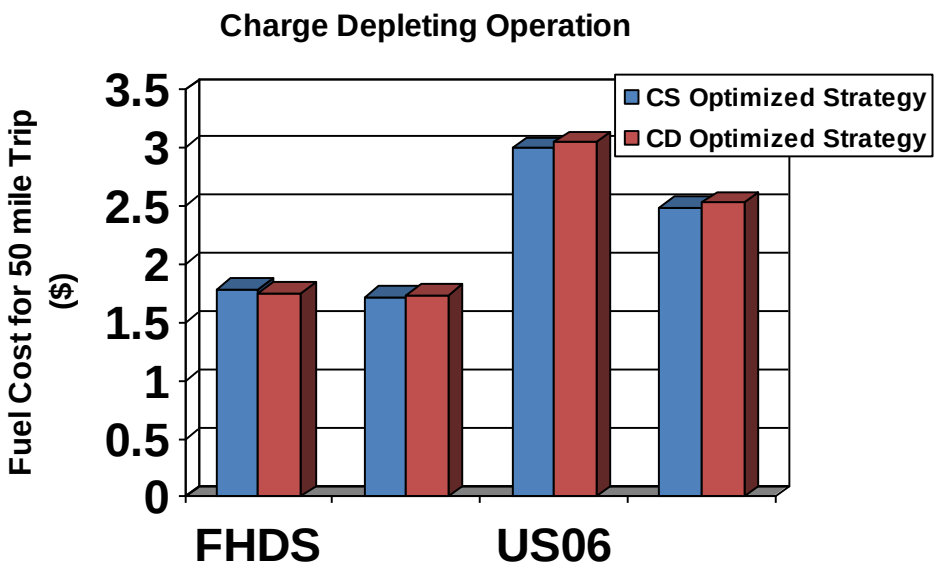


Vehicle Fuel Economy – Torque Control

- Comparison to Ideal Operating Line Strategy
- Identical Engine Pull-up Controllers

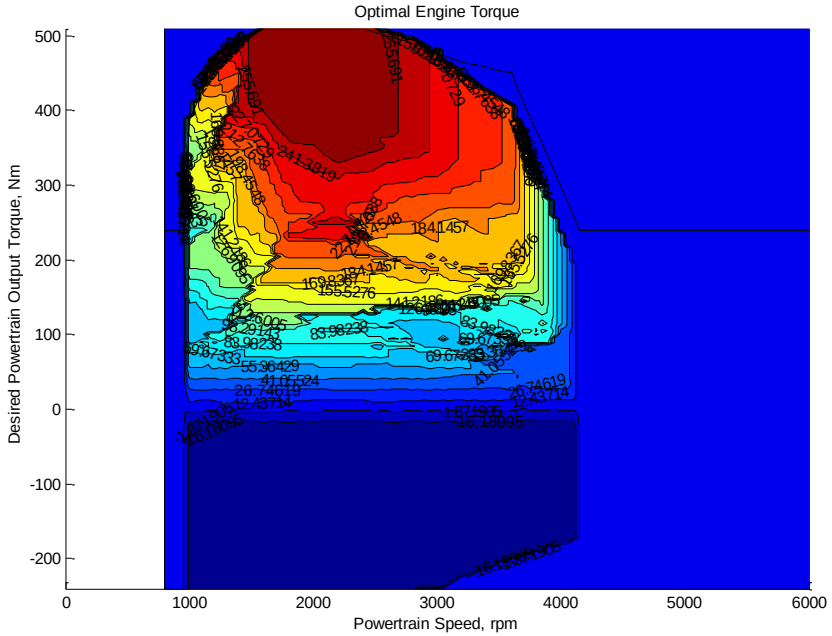




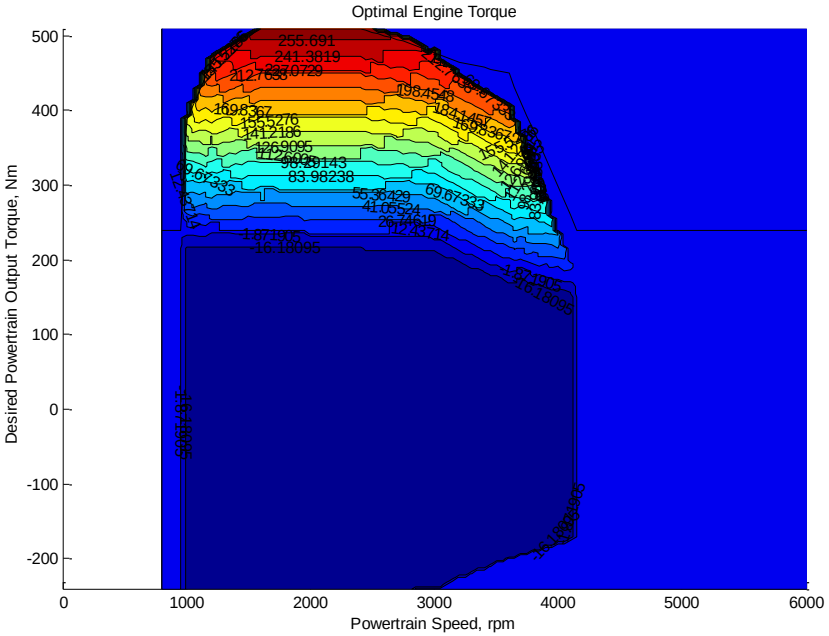


Vehicle Fuel Economy – Torque Control

- Results of Optimization
  - *T_engine = f(rpm, T_command)*



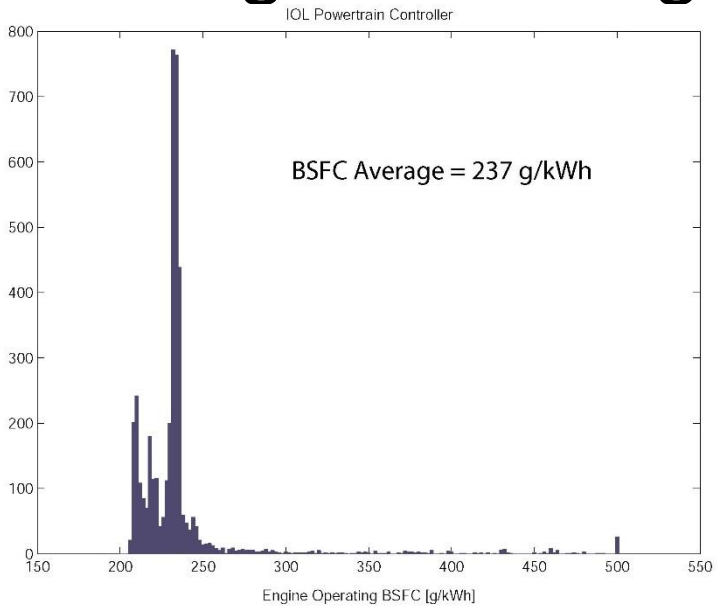
Battery Sustaining

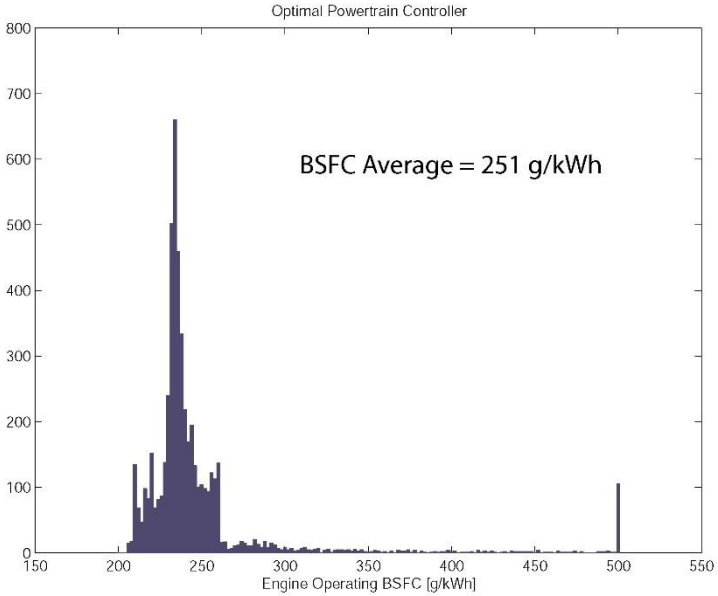


Battery Depleting – Fueling Cost Optimized

Vehicle Fuel Economy – Torque Control

- Charge Sustaining FUDS Cycle Testing





- Relatively Uncompromised System Efficiency Improvement by System Optimization

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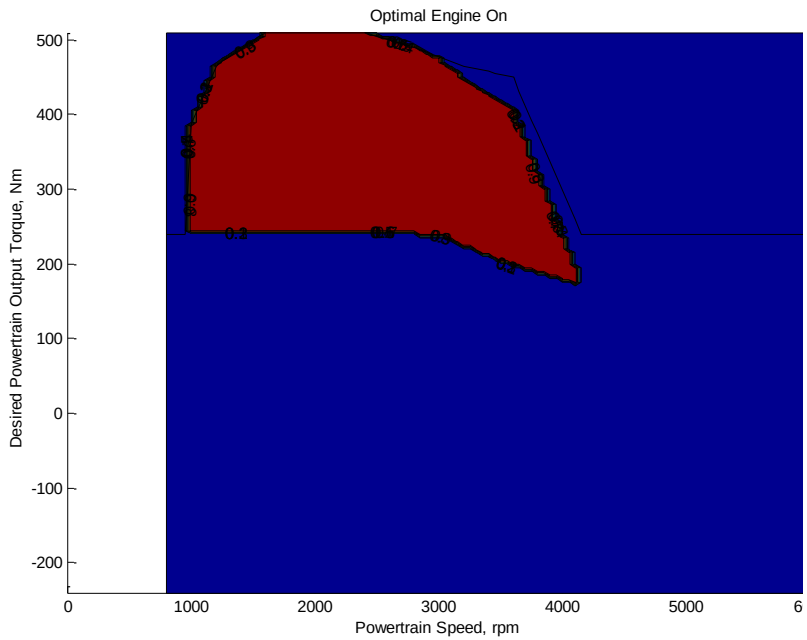
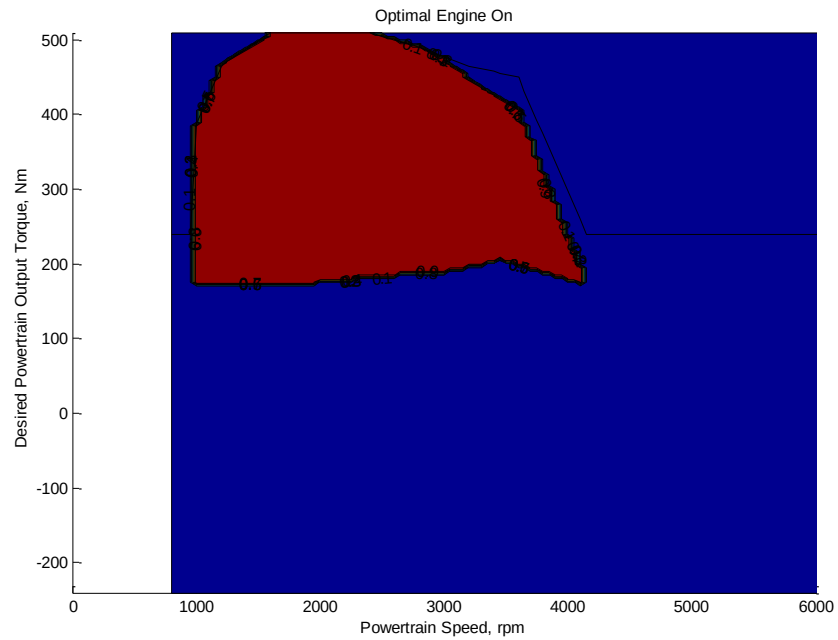
- Using a similar evaluation technique one can consider whether the engine should be on.
  - *Engine_On_Cmd = f(rpm, T_command)*



Battery Depleting – Fueling Cost Optimized



Battery Sustaining

- For Engine On/Off Control there are other considerations/compromises to be made